Project Goals

Deliverables, What we want to do, What is due

Week1: N/A

Week 2: N/A

Week 3: First Meetings

- Learning Softball & The Dataset
    - Softball learning the acronyms that are used in the dataset

Week 4: What to do?

- Learning Softball & The Dataset
- Understand what we can do with this data
- Make goals and set down what questions we have
    - Find our own data, even if it is already shown in visuals or words, for the coaches
    - What can we build and what can we do to make a long-term impact
    - How are we going to do that

Week 5: Our Thoughts

Project Objective

The goal of this project we want to pitch this week is to build a data-driven decision support system for Purdue Softball. The system will analyze game situations such as do-or-die baserunning and sacrifice fly scenarios by measuring timing, player movement, and team-level data to identify areas for improvement during games.

In parallel, the project will replace manual paper based stat entry with an automated digital workflow. This includes building a lightweight web or tablet-friendly application connected to an internal API that allows coaching staff to enter player and game data directly from an iPad.

The team will execute the project through incremental weekly deliverables, building toward a complete working prototype by the end of the academic year.

Idea For US

- Pull the CSV and pull the data from the other team our team, make it coach friendly for now, this is from trackman
    - We can analyze the data and make dashboards and visuals that the coaches can use during the game whether that is pitching or for scouting from the other team
    - We can use AI and scope for the project as long as it is a tool to build off of
    - We can use the one drive to mess around with the data

Clean Project Questions

Data Optimization / Game Decision Questions

- What measurable timing factors influence success in do or die base-running situations? How many seconds are available between ball contact and scoring opportunity in sacrifice fly scenarios?
    - Look at the grouping for the practice day then analyze from there by player
    - Do or die is on the ground in the outfield
    - Always less than 3 seconds for sac fly to get them out
    - Do or Die is around 5 seconds is optimal
    - Faster the better
- Can a model recommend tactical adjustments based on historical timing data?
    - Use PowerBI and look into what we can do with the past data to make it useful for the future games
- How can insights be presented to coaches in a clear and actionable format?
    - Format to the coaches use whatever software we feel is necessary

Data Collection / Automation Questions

- What information is currently recorded manually on paper?
    - Our stats as in pitching to the other team and we are on defense
    - Also the stats of when we are hitting
- What is the simplest interface that allows fast entry during live games for data collection?

    - Figuring that out if Kate can use IPad or else no external technology is allowed
- How should data be validated and stored through an internal API?
    - We can figure out as we go on with more research
- What database structure best supports fast retrieval and analysis?
    - We will do research to find out what is the best one

System Implementation Questions

- Should the interface be a progressive web app or native tablet app?
    - When Kate gets confirmation if we can use one for the games then we will decide
- What authentication or access controls are needed?
    - Trackman and fileZIla
    - PowerBI Pro potentially
- How will the system scale if expanded to other teams or seasons?
    - We will figuring that one out as we go

Big Picture: One Big Website where we can find all the data and trends it can pull the data from one drive where everything is in house then auto make the diagram and visuals.

IP Meeting Notes

Meeting 1 Notes (1/27/26) Tuesday

Points of Contact

Kate Claypool - Softball Director of Player Development claypook@purdue.edu
Magali Nadia Frezzotti - Softball Head Coach mnfrezzo@purdue.edu

Assignment 1: Sharing Communications, who will schedule meetings, share questions, and essentially work as the head communicator. Due 1/29/26 Shubh (most likely)

Times to Meet 8:30AM-9:30AM In Person every Tuesday

Notes: Kate on what data we have

Offensive Databases

- Hitrax is used for the hitting data for offense, collecting outgoing ball flying metrics
- Challenge: manually export each csv for each player for each session
  Has all data, but not initially an offensive data database without sorting what we want it to be is an automated dashboard
- Goes back for the fall of 24 Excel doc **manual data inputs, but want to automate?**

Defensive Database

- Internal data collection: Defense catchers' pop time, over are vlow, timing metrics,
- Pitching uses Rapsodo CSV export, whole session, to different databases for pitching that is on performance metrics

Questions we want to answer for our project for now

- We want to use data to know when to make changes in the defense (maybe a Pichter change)
- Is it useful to share all the data, or do we want to group it for different projects?

Questions I have

- Q: How can we make offensive changes, is it like setting the battering order against different times or just for practices to focus on certain things?
- Ans: Predict or have a better game strategy or a better option
- Q: How do we make the defense more specific? Can we change the field or outfielders to different positions?

- Q: What does the pitcher data look like? What are we looking at, and how can we make informed decisions?
- Ans: All pitching performance is one of them, then one is the leverage side. Leverage chart when we were in the situation the most and what happened. The other side is all the other data

Timeline of Softball

- Start games next week
- The first 4 weeks of working from home/close games
- Tuesday and Wednesday are usually practice days in the afternoon
- Games they leave Thursday come back Sunday
- Big ten games: one opponent plays 3-4 times, the schedule depends on whether it's away or home

Meeting 2 Notes (1/30/26) Friday

Defensive Data

Defense Catching Measurables Excel

- **Pop time** throw from the catcher's glove to the second base glove
    - Game, drills, practice: data is aimed at 1.6 for best, average is 1.7-1.8

Defense In-Practice Measurables Excel

- Game Winning, Do or Dies, Sac Fly-OF is game-winning data
    - Sac Fly-OF: is in practice where it is measured, and the times can vary because the distance can vary, but always in the outfield
    - Do or Die: measured only in practices. Hit to the outfield, but on the ground, times can vary because of the distance hit and the distance they have to throw
    - Distance isn't recorded, but something that can maybe be done

Outfield Measurables Excel

- The team as a whole, for Sac Fly-OF and Do or Die

Overhand Velo Testing Excel

- Velo is throwing power
    - Top 10% is green, the bottom 10% is red
    - Miles per hour, how hard they can throw the ball from 30 feet away
    - 30 feet from regular anyway they want, then the ground ball is from their feet, a fly ball from catching in the air

Offensive Data

Run Expectancy Chart Excel | 643charts | Bunting strategy chart

- Last season, game metrics of the situation of the number of people on base
- Win percentage based on bases and outs
- Project potential: Making the workflow automated, automatically put into the database.

Spring 2026 Offensive Stats Excel

- What the data acronyms mean
    - AB: At Bat
    - Strikes: in the zone misses swing or doesn't, and gets called, swing makes a foul (unless 2 strikes already),
    - Balls: outside the zone, the hitter doesn't swing
    - BB: Base on balls
    - S/M: Swing and Miss
    - K take: Strike taken
    - Foul: Foul Ball

- Count: Count of the situation that resulted in the event
- 1P SW
- Hit/Pit/Even:
- 1B: Single
- LD: Line drive
- LO: Line out
- GP: Ground Ball
- BACKK: strike no swinging
- RBI: Base runner scoring based on what the hitter does
- Moved Runner: Did someone move if the ball was hit
- QAB: Quality at bat, if you move a runner lots of ways to show something happened
- BA: batting average. Did you get on base or score
- OBP: Did you reach base successfully
- BABIP: ball average balls in play

Meeting 3 Notes (2/03/26) Friday

Other Folder

Keys and Definitions Folder: Has a lot of definitions and keys for the acronyms

D1 Runs Per Data 2020-2025

- All the runs per earning for teams

D1 Softball Run Expectancy Chart pdf - (6-4-3 charts)

- Stats of player performance and team performance
- Pitching Stats
    - 1-1%W% who wins after a one, one count
    - FIP: Pitcher's ability to produce outs
- Changes with this data to maximize player performance against the best statistical output

Pitching Data

Ogx Softball website

- Uses data from practices, RapSodo 2.0, and Trackman does the calculations, and then manual input for which type of pitch being thrown
- Pitches are seen as a clock in terms of the quadrant, the quality of the pitch
- It is tracked on which part of the zone, there can be a vertical break or a horizontal break, depending on the type of pitch
    - Based on pitch profile and matchup, when they go in or not

Pitcher Stats Fall 2025 Excel

- This data is on scrimmages and match-up data
- Hitters are categorized in 4 sections: lefty or righty, hitter type: fly-middle-ground,

Meeting 4 Notes (2/10/26) Tuesday

Starting Notes - Questions, General Game

- Focus on data with the bigger picture but as we go on learn more about the small nuisances of softball
- First number for ump is always balls second number is the strikes

Meeting 5 Notes (2/17/26) Tuesday

Is the Database Clear? Article

- Read an article about how pros vs college and noticed there is a big difference
- Potential to create internal database
- We are a team of analytics and we can do a lot
- Data is evolving from the college level

Presentation Notes

- Kate will get back with more notes
- Overall use Rapsodo or trackman for more intensive data research
    - It is hard to tell which one is correct but try to find the correct answer
- We are good to go once we get approvals from different items
- Make sure we have a more clear timeline and note that if something comes up we can highlight it

App for Inputting Data Update

- Confusing answer but it is looking more likely we can do this
- Essentially to enter information that would feed into an csv automatically but not do anything else
- We will work towards creating this Ipad in project 2

Next Week Deliverable

- Try to create a model for the data cleaning csvs
- Create a basic example of what the csv cleaned looks like for hitrax, trackman, and rapsodo
- Goal: Create an internal

Meeting 6 Notes (2/24/26) Tuesday

Shiny App Notes - pulling trackman file where can we go?

- Have game by game usefulness
- Have a collection of pitcher overviews from Purdue's team
- Date Range from other teams using their game data
    - Apply by date range for prediction and scouting purpose
    - Apply to see how they change
    - Apply for different situation

- Comparing game to game from live pulls
    - Using one pitcher saw dropballs, but then maybe next time to switch it up to throw high (app can make that suggestion)
- Interesting to see if we can clean the csv dropdown (the name)
- Fix the names of the files, make it the opponent vs who they played, and the date
- UI Design
    - Drop down by data type (FTP, HitTrax, or Rapsodo)
    - Drop down by team name
    - Drop down by type of game (live, bullpen, scrimmage)
    - Drop down by pitcher / batter and their handness
    - Drop down by day or day range

- Visuals
    - Pitch Usage %
    - Pitch Movement
    - Pitch Location
    - Built Strike Percentage → open to any other insights for scouting purpose
    - Based on the current table → add average vertical / horizontal break (prediction purpose: how their pitch changed over times)

Cleaned the data so it is easier to see and check
Really focus on the pitcher dataRight handed vs left handed hitter

Pitcher vs them, how they preform

Can we isolate best match up

HitTrax Notes - what can we do that the app already doesn't?

- Simple metrics, but try to focus on perfect areas that can be better for our hitters
    - Appends CSV and gives metrics, but it can be changed for what we are looking for

- Up-to-date long-term swing analysis
- Maybe use AWS for the cloud case
- How can we use the app outside of what hittrax already does
    - Overall report from the hitter
- Continue to expand on what the hitter does, maybe pull trackman data or notes

Overall Notes

- Really like the direction this is headed
    - Has other ideas but we will wait and see where this is going first
- Will follow up with us if there are any other notes on what we can do
- Look into automation, but it will take time

Meeting 7 Notes (3/02/26) Tuesday

Shiny App Notes - Kate thoughts on new updates

- How do we optimize what we have available
    - How do we produce outs and ground balls double plays
    - Same category of pitcher how can we build a slow change to help our pitchers
- Notes on website
    - We don't need the avg speed
    - We can compare Purdue vs how other do
    - Ease of use set it as a predetermined use for Purdue
    - OGX Softball dashboard we can show more of what they do
        - How has the pitch been developed over time
        - Hard to look at the scope particularly
    - Rapsodo will cloud sime areas of data usage
- This is something that should be futured proofed not just for this season
    - We need to be able to have across the board usable
    - Old batter or hitter that can be a bullpen hitter
- We can pull the old docs on the bullpen and batter practice to look for key identifies
- Questions for Kates Answers
    - From the start show the current spring season and Purdue
        - Have the option of the date range still
        - Team name is the best, if there is a team we need to look at or continue to look at
    - Strike percentage is the total piches, swings and misses in the whiff percentages
    - Batter visual: Let's see metrics on swing decisions
        - like how often chase percentage: how often they swing on a pitch outside the box
        - Are they aggressive: do they always swing on any pitch
    - Keep UI the same for now and once we get more into it we can start adding the new metrics
- Post game CSV we want to produce a PDF of what we have for Shiny App
    - We get the pitching report for our team and the opponents team
        - Strike percetenage
- Focus on Rapsodo right now on Shiny App
    - Won't have whiff%
    - Same thing same page show tab for trackman only, rapsodo only, combined
    - One pitcher per page, one pdf for one team and one for our team

Meeting 8 Notes (3/10/26) Tuesday

Shiny App Notes - Kate thoughts on new updates

- Summary table notes
    - In game of live game metrics, what is it pulling from
    - If we do summary type is all, we should make it still separated
    - Leave undefined off the pitch type
- Comparison tab notes
    - Change up off off speed or change up usage
    - Get rid off spin rate
    - Have a comparison between Purdue Pitchers
- We can have the all tab broken into sections but it has the same type of info as the main tab just different sections
- We could preload to the backend with the type of CSV before putting it in
- For what to give the coaches it will be both will get the pdf and Shiny App
    - Have an in game analysis of chase data etc
- Use the OGX Softball as a blueprint of what we want do with the pitchers game y game analysis
    - Look at their summary tab like overall, pitch type, and batter side is going to be useful for opposing pitchers
- Look at the same thing that the same summary tab but now for our hitters
    - What pitcher left or right what looks different in terms of the outcome
- Be able to look at this for more of live game situations, since the data is not that large we can look at which pitches the other team hitters are struggling with and vice versa which ones they are doing really good in
- Rapsodo focus on jiro and other metrics not the spin efficiency

Meeting 9 Notes (3/24/26) Tuesday

Pitcher Profile Notes

- Create pitcher right or left hand batter against pitcher analysis
    - For example how does a pitcher do vs a right handed hitter vs left
- Create an average pitch movement average or std to show a range instead of each individual pitch
    - Create a ratio for the pitches thrown for the pitch location and movements

Final Dilverable Notes

- Continue to work on ShinyApp and PDF integration but make this able to be future proofed
- Trying to make this a tool that can be used in the future
- Create a readme report of everything we have done including code, writing, etc
- We want to be able to work on three main things to come together rather than starting a new project

What to do this week

- Create a new pitcher report for the pdfs by thursday
    - Activily using it right now in games so that is nice
- Try to make the scouting coach side more friendly to them for the pdfs
    - They are already using it and the like it so far!
- Be able to make all this data look friendly and more manageable
- Come back next week with a predictive component with what we already have

Meeting 10 Notes (3/31/26) Tuesday

Shiny App/PDF Small Integrations

- Create a zoom in to graph only application for the different graphs to fit the screen
- Adding a search bar for the team names
- Keep the PDF as the same for pitches don’t need to cluster

Predictive Analytics Tab

- Keep working on this type of model and validate the information until there is a high confidence level
- Left handed batter vs right handed batter

Meeting 11 Notes (4/07/26) Tuesday

Predictive Analytics Tab

- Keep working on this type of model and validate the information until there is a high confidence level
- Left handed batter vs right handed batter is now in place and same with pitchers (there isn't many pitchers that are left handed)
- XGBoost filters and testing different variables to keep it updated with new analysis

Next Steps for Project

- How the coaches what they want then show them how to use it how to use it
- Goal is to explain how the tabs and data works
- Give them live demonstrations of what we have and a summary over what we do and what is done
- Rapsodo integration could be a possibility
- Create the framework of how to clean and hope to work on the data
-

Meeting 12 Final Presentation

Meeting 13 Notes (4/21/26) Tuesday

ShinyApp Tab

- Keep the verified vs unverified tabs but we will need to look the options if there is barely any pitchers what to do

Predictive Analytics

- Adjust the wordage ex: ride to something else for the explainability
- Overall college level for top 75
- League wide data integration and tweaks that we needed to be done and logic is the same
- Making a change from control impact vs overall pitch impact to see if there is any changes
- Make a key of what the Impacts mean and if + or - is good or bad

What to do from here

- Hand off material to Kate

Team Meeting Notes

Meeting 1 Notes (2/12/26) Friday

Agenda

Gary Meeting

- Find a time to meet to discuss the Following with him
    - Team participation during client meetings: entire team must participate whether asking questions, leading the presentation, etc.
    - Internal review process before sharing materials externally
    - Weekly deliverables must reflect preparation and professionalism
        - Times: Monday 9am & 10am,3pm, Tuesday: 9:30am to 4:30pm besides 2:30pm, Wednesday: 1:30 &2:30pm

What Research needs to be done

- What has everyone done to learn more about the game
- What has everyone done to learn about what to implement for our project
- Can we focus on getting work done week by week as a group or should we split up
- Do more research on FTP data file and decide what important and can be included in the dashboard → Divide it into pitching and offense
- What can deliver for Tuesday (2/17) Meeting

Next Steps

- Focus on data implementation
- Need to work on the final deliverable is in scope and we agree on it as a team
- What can we do better

Meeting Deliverable with Kate (2/17): 3 projects

- Big picture dashboard project
    - Info teams want to include (focus on offense and pitching)
        - Graph: pitch location, pitch movement, pitch usage % → what column from FPT(trackman) file created for these visuals
        - Pitching: drop-down (pitch type, velo, etc)
    - Info Kate and Coach want to include
    - Connect excel file for extracting intel
- An app (hold-up on this)

Meeting 2 Notes (2/19/26) Friday

Agenda

Gary Meeting - every 2 weeks

- Updates we need to tackle
    - Meet twice a week
    - Contact him about our deliverables

Cursor/VS Code Sync with GitHub

- Was everyone able to get it setup/learn how it works
- Does anyone need help syncing anything up
- Cursor/VS Code IDE understandable?
- Was everyone able to download all the data and sign into everything
    - Download from HitTrax?
    - Download from Rapsodo?
    - Download from 643Chart?
    - Download from Trackman?
- Conduct some test sampling on what data is important (waiting on Kate but we should continue with what we believe is important)
    - FTP: what should be included
        - Game Detail:
            - Date
            - Inning
            - Top/Bottom
            - Outs
            - Balls
            - Strikes
            - PAofInning
            - PitchofPA
        - Identifier
            - Pitcher Name
            - PitcherID
            - Batter Name
            - Batter ID
            - GameID or GameUID (for joining purpose)
            - Pitcher / Batter Team
        - Handedness
            - Pitch Throws
            - Batter Side
        - Pitch Classification

            - Tagged Pitch Type
            - Pitch Call
            - Tagged Hit Type
            - Play Result
            - Out On Play
            - Runs Score
        - Pitch Quality and Movement (check in with everyone)
        - Pitch Location (Kate mentioned want it on visuals)
            - Plate Loc Height
            - Plate Loc Size
            - Zone Speed
            - Zone Time
            - Can be considered: pfxx / pfxz
- Create basic CSV outputs of cleansed data
- Show some Power BI outputs of Visuals

IP Brightspace Tasks

- Go over the doc
- Tasks are due soon
    - CrossRoad Classic Analytics Challenged
    - Organized in one folder (Purdue Template)
        - Slide decks
        - Code
        - Data
    - Slide decks for “Business Problem”
    - Slide decks for “Business Benefits”
    - Slide decks for “Business to Analytics Problem Framing”
    - Use Smartsheets for tracking purpose
    - Point of Contact person summarize and post on Teams

Next Steps

- We need to find a good schedule to do work
- Need to create weekly slides on what we have done
- Anything we should change or do differently now is the time to address it

Meeting 3 Notes (2/26/26) Friday

Agenda

- Finish and organize the weekly slides
    - Update new data on Shiny
    - Update New Data on HitTrax
    - General updates
    - Questions
- Continue to show progress on the Shiny App on the slides
    - Show the new Iterations Kate wanted
    - Show the cleaned UI, new metrics, etc
- Continue to show progress on HitTrax work on the slides
    - Show the new Iterations Kate wanted
    - Show the more insightful analytics, the app doesn't already show
- Complete Task 2 before Tuesday - who can tackle this?
- Schedule the next meeting with Gary and Kate
    - Setting up another meeting w Gary
- Start planning around home games beginning March 6
    - Who can go whos here? - Shubh and Tam (Saad tbd)
    - Tell Kate to get Access Pass
- Stay aligned on first-pitch details and related communication
    - Shubh is currently leading the communication for this
    - Right now we plan on the second 2 dates, April 2,4
        - Cool Opportunity
- Find a new Day to meet
    - Need to find new dates based on the new module schedule to meet
        - Saad schedule - thurs, friday sat, sun
        - Shubh schedule same schedule
        - Tam same schedule
        - Wenesday work day, Friday updates notes before me meet with Kate
            - 6-7PM on Wenesday
            - 12PM on Friday
            - Tuesday meet w Kate 8:30

Shiny App Next Steps

- Continue improving the app so it is useful on a game-by-game basis

- Build pitcher overview pages for Purdue players
- Add date-range filtering for scouting, prediction, and trend analysis
- Support game-to-game comparisons using live pull data
- Explore ways the app can suggest pitch adjustments based on prior patterns
- Clean the CSV dropdown labels so names are easier to read and select
- Standardize file naming to include opponent, matchup, and date
- Finish core UI dropdowns for:
    - data type (FTP, HitTrax, Rapsodo)
    - team name
    - game type (live, bullpen, scrimmage)
    - pitcher or batter with handedness
    - day or day range
- Prioritize visuals for:
    - pitch usage %
    - pitch movement
    - pitch location
    - strike percentage (strike Zone)
        - Value: StrikeCalled, StrikeSwinging, FoulBallFiedable, FouldBallNotFieldable, Inplay
        - Divided by total pitches
- Add average vertical and horizontal break to support prediction and scouting
- Keep cleaning the data so it is easier to read, filter, and validate
- Focus more deeply on pitcher data, especially:
    - right-handed vs left-handed hitter matchups
    - pitcher performance by matchup
    - identifying best matchup opportunities

HitTrax Next Steps

- Define what value we can add beyond what HitTrax already provides
- Focus on simple but useful metrics that better identify ideal areas for hitter improvement
- Adjust current CSV append outputs and metrics to better match team needs
- Build toward up-to-date, long-term swing analysis
- Explore using AWS if cloud storage or processing becomes useful
- Expand hitter reporting beyond current app outputs
- Consider pulling in Trackman data or notes to strengthen hitter analysis

Project-Wide Next Steps

- Keep building in the current direction since feedback has been positive

- Stay flexible and wait for additional ideas before expanding scope too quickly
- Watch for follow-up notes and feedback from the team
- Begin thinking about automation opportunities, while recognizing this will be a longer-term step

Shinny App Question

1. 2 main panel:
    a. Home Page
        i. Team performance in general
            1. Win / loss percentage
            2.
    b. Player Performance